

Anurag Saurav Mathew

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🌐 LinkedIn/Anurag-Mathew

🔗 Developer-Portfolio

EDUCATION

BSc (Honours) Computer Science | University of Manchester

Sep 2021 – Jul 2025

- **Achievements & Activities:** Graduated with honours being placed in the Second class, Division one (2:1 classification), First Class in User Experience module, Developed an **AI agent for Hex** with a 74.1% win rate and ranked **19th** in speed (78.6 mark) among all submissions, Integrated **Spotify API** into '**GhostTyper**' for real-time music playback; Optimised database efficiency with MySQL indexing and caching, improving query speed by **30%**. Enhanced user engagement by dynamically adjusting difficulty based on lyric tempo.

Higher Secondary Certificate, Class XII | Shubham Raje Jr. College, Thane, India

2021

- **Marks:** 97.5% (Mathematics & Statistics: 99%; Computer Science: 97%)

Indian Certificate of Secondary Education, Class X | HFS, Thane, India

2019

- **Marks:** 94% (Mathematics: 97%; Computer Science: 100%; Science: 94%)

WORK EXPERIENCE

Technical Support Engineer | Beckhoff Automation UK Limited

Oct 2025 – Present

- Provide first-line technical support across **TwinCAT 3 PLC/HMI**, **EtherCAT** networks, **Beckhoff IPCs**, I/O and motion; triaging tickets, reproducing issues, and delivering fixes/workarounds.
- Build customer-facing **demo projects & proof-of-concepts** to accelerate adoption and de-risk integrations.
- Deploy and configure advanced motion control applications for **XTS** and **XPlanar** magnetic levitation systems using **TF5000**, supporting customer commissioning and integration workflows.
- Delivered on-site technical support and consultancy visits to high-profile customers including **Mercedes AMG High Performance Powertrains** (Brixworth), **Package Automation** and **TrayTec** (food packaging industry, Manchester), and **Pressure Designs** (Sheffield).
- Co-delivered a **TwinCAT PLC training course & TwinCAT HMI training course** at Alderley Park, Beckhoff UK, covering TwinCAT 3 fundamentals, structured text programming, HMI Creation and IEC 61131-3 best practices.

Food & Beverage Team Leader | Stockport County FC

Feb 2024 – Sep 2025

Managed bar and retail operations during matchdays, supervising teams of up to 10 staff in high-volume service environment.

Front of House Team Leader | Manchester United FC

Jan 2022 – Jun 2025

Led teams of 5–8 staff in high-pressure service areas, ensuring smooth operations and fast turnaround during peak demand.

PROJECTS

Industrial Automation PLC + HMI | Beckhoff RT Linux

Dec 2025 – Present

Developed automated deployment scripts and engineered a complete PLC solution with integrated HMI on the CX9240 embedded controller running Beckhoff RT Linux. Implemented motion control, I/O processing, and EtherCAT diagnostics using TwinCAT 3 with object-oriented programming principles. Built a lightweight HMI with responsive navigation, real-time trending, and historization capabilities. Additionally developed the **Linux IPC Setup Console**, a **Electron** desktop application that turns the full CX9240 provisioning and management workflow into a GUI. The app connects to a target IPC via SSH, authenticates with myBeckhoff credentials, and automates: TwinCAT runtime installation, optional package management with **version pinning**, APT auth configuration, HMI server initialisation, and **TF1200** kiosk/Chromium setup via a point-and-click JSON config editor. A dedicated **CX Management** module handles network interface configuration (DHCP/static), nftables firewall rule management, and selective package upgrades. Features include a **live SSH terminal** streaming real-time output, MyBeckhoff credential validation, a *Fetch From Feed* function that queries the target's apt cache to auto-populate available packages, and timestamped config backups before any writes. Distributed as a portable

Windows .exe requiring no installation or Linux expertise.

Walking on Words (Final Year Dissertation)

Sep 2024 – May 2025

Formalised a novel "walking" model of word generation, extending primitive-word theory to bounded-step surjective walks. Designed and implemented algorithms to explore combinatorial properties, achieving efficient $O(n^4)$ solutions and validated results via experimental evaluation. Commended by examiners for depth and rigor.

Formula Student AI Developer | Manchester Stinger Racing Society

Sep 2023 – Jun 2024

Headed a team of **5** developers in designing and implementing the first AI iteration for the society's autonomous racing vehicle, achieving milestone completion ahead of schedule.

Full-Stack Developer | UniCS Development Team

Sep 2023 – Jun 2024

Designed, built, and maintained an internal events database and public-facing website used by **1,000+** students for society event management. Implemented responsive front-end features and optimised backend performance, reducing page load times by approximately **35%**. Coordinated with multiple society committees to gather requirements, prioritise features, and deploy updates in line with event calendars.

TECHNICAL SKILLS & CERTIFICATIONS

- **Software Development:** MERN Stack, Tailwind, Bootstrap, JavaScript, Linux, Next.js, Django, Spring, PHP, Electron
- **PLC Programming:** TwinCAT 3, TwinCAT HMI, TwinCAT Motion, Beckhoff RT Linux, IEC 61131-3
- **Data Science & ML:** Dash, scikit-learn, TensorFlow, PyTorch, Transformers (Hugging Face), OpenCV, SpaCy, SciPy
- **Others:** Go, Python, Java, C, C#, C++, React, Haskell, Leaflet.js, REST API, ARM Assembly, Verilog, SQL, Docker, AWS, HTML, Experienced with GitLab & Github, large codebases, debugging, and agile development
- **Certifications:** **Software Engineering - JP Morgan & Chase**, TwinCAT PLC, TwinCAT OOP, TwinCAT Motion, TwinCAT HMI, TwinSAFE